IES FRANCISCO DE GOYA

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CURSO: DAM 2

Programación Multimedia y Dispositivos Móviles: Como calcular propina

A continuación se presentarán evidencias sobre el ejercicio planteado en el aula virtual, se presentarán tanto las imágenes como el texto plano del fragmento del código:

MainActivity.kt

```
package com.example.changebgcolor

import android.app.Activity
import android.content.Intent
import android.graphics.Color
import android.os.Bundle
import androidx.activity.result.contract.ActivityResultContracts
import androidx.appcompat.app.AppCompatActivity
import com.example.changebgcolor.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity() {

    private lateinit var binding: ActivityMainBinding

    // Registrar el contrato para obtener el resultado de la segunda
actividad
    private val colorPickerLauncher = registerForActivityResult(
        ActivityResultContracts.StartActivityForResult()
    ) { result ->
        if (result.resultCode == Activity.RESULT_OK) {
            val data: Intent? = result.data
            val colorHex = data?.getStringExtra("COLOR_HEX")
            val colorName = data?.getStringExtra("COLOR_NAME")

            if (colorHex != null && colorName != null) {
                // Aplicar el color de fondo al layout principal

binding.mainLayout.setBackgroundColor(Color.parseColor(colorHex))
                // Actualizar el texto indicando el color seleccionado
                binding.tvCurrentColor.text = "Color actual: $colorName"
            }
        }
        // Si se cancela o se pulsa atrás, no hace nada y mantiene el color
actual sin errores.
    }

    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
        binding = ActivityMainBinding.inflate(layoutInflater)
        setContentView(binding.root)

        binding.btnChooseColor.setOnClickListener {
            val intent = Intent(this, ColorPickerActivity::class.java)
            colorPickerLauncher.launch(intent)
        }
    }
}
```

ColorPockerActivity.kt

```
package com.example.changebgcolor

import android.app.Activity
import android.content.Intent
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import com.example.changebgcolor.databinding.ActivityColorPickerBinding

class ColorPickerActivity : AppCompatActivity() {

    private lateinit var binding: ActivityColorPickerBinding

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        binding = ActivityColorPickerBinding.inflate(layoutInflater)
        setContentView(binding.root)

        supportActionBar?.setDisplayHomeAsUpEnabled(true)
        title = "Elegir Color"

        // Botón Rojo
        binding.btnRed.setOnClickListener {
            devolverColor("#F44336", "Rojo")
        }

        // Botón Verde
        binding.btnGreen.setOnClickListener {
            devolverColor("#4CAF50", "Verde")
        }

        // Botón Azul
        binding.btnBlue.setOnClickListener {
            devolverColor("#2196F3", "Azul")
        }
    }

    private fun devolverColor(hexCode: String, colorName: String) {
        val resultIntent = Intent().apply {
```

```
            putExtra("COLOR_HEX", hexCode)
            putExtra("COLOR_NAME", colorName)
        }
        setResult(Activity.RESULT_OK, resultIntent)
        finish() // Cierra la actividad y regresa a MainActivity
    }

    override fun onSupportNavigateUp(): Boolean {
        onBackPressedDispatcher.onBackPressed()
        return true
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/mainLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    android:background="@android:color/white">

    <TextView
        android:id="@+id/tvCurrentColor"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Color actual: Predeterminado"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginBottom="32dp"
        android:textColor="@android:color/black" />

    <Button
        android:id="@+id/btnChooseColor"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Elegir Color de Fondo"
        android:padding="16dp" />

</LinearLayout>
```

activity_color_picker.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selecciona un Color"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="32dp" />

    <Button
        android:id="@+id/btnRed"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Rojo"
        android:backgroundTint="#F44336"
        android:textColor="@android:color/white"
        android:padding="16dp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnGreen"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Verde"
        android:backgroundTint="#4CAF50"
        android:textColor="@android:color/white"
        android:padding="16dp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnBlue"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Azul"
        android:backgroundTint="#2196F3"
        android:textColor="@android:color/white"
        android:padding="16dp" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
```

```
        android:icon="@mipmap/ic_launcher"
        android:label="Cambiafondos"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Material3.DayNight">

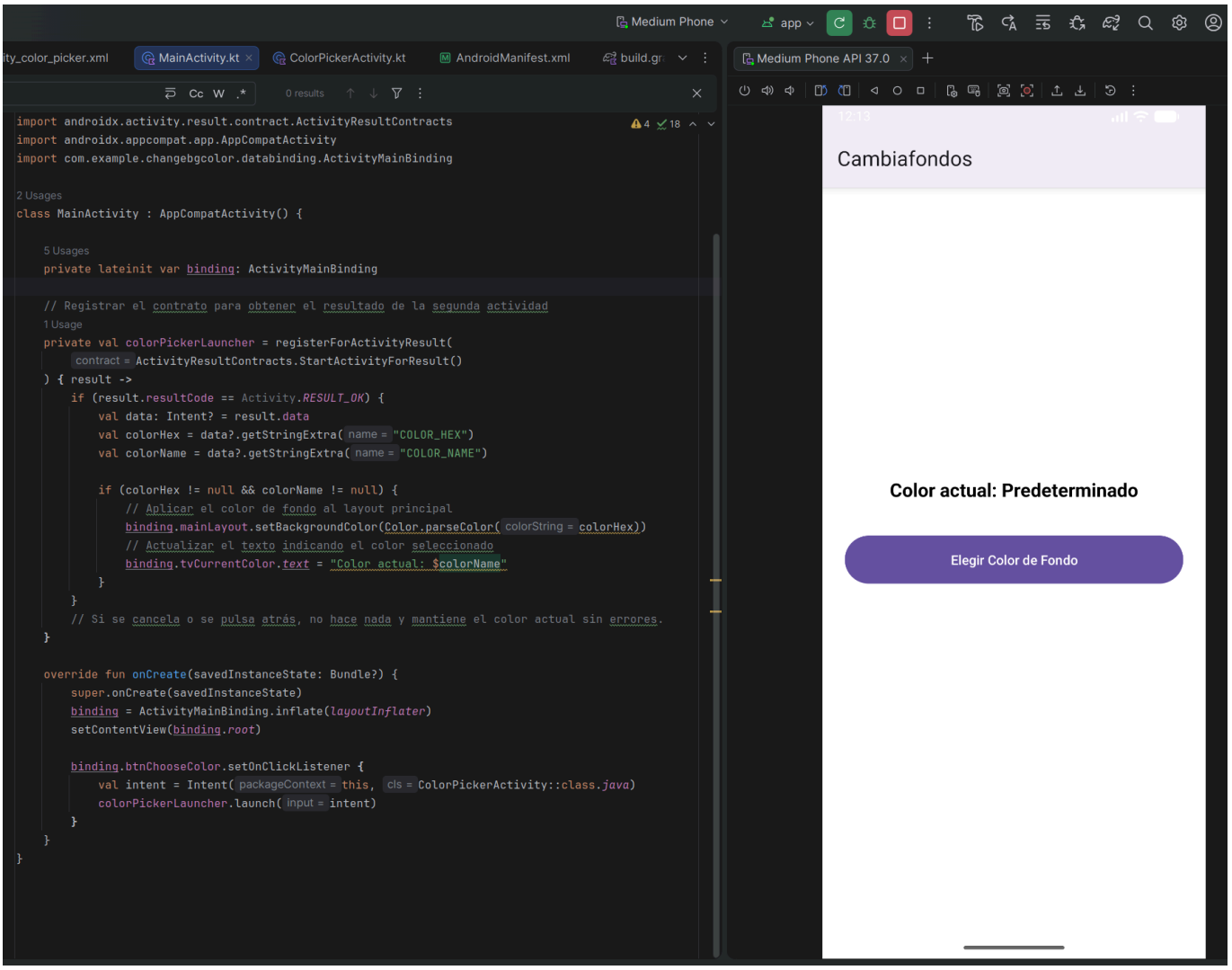
        <!-- Actividad Principal -->
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER"
/>
            </intent-filter>
        </activity>

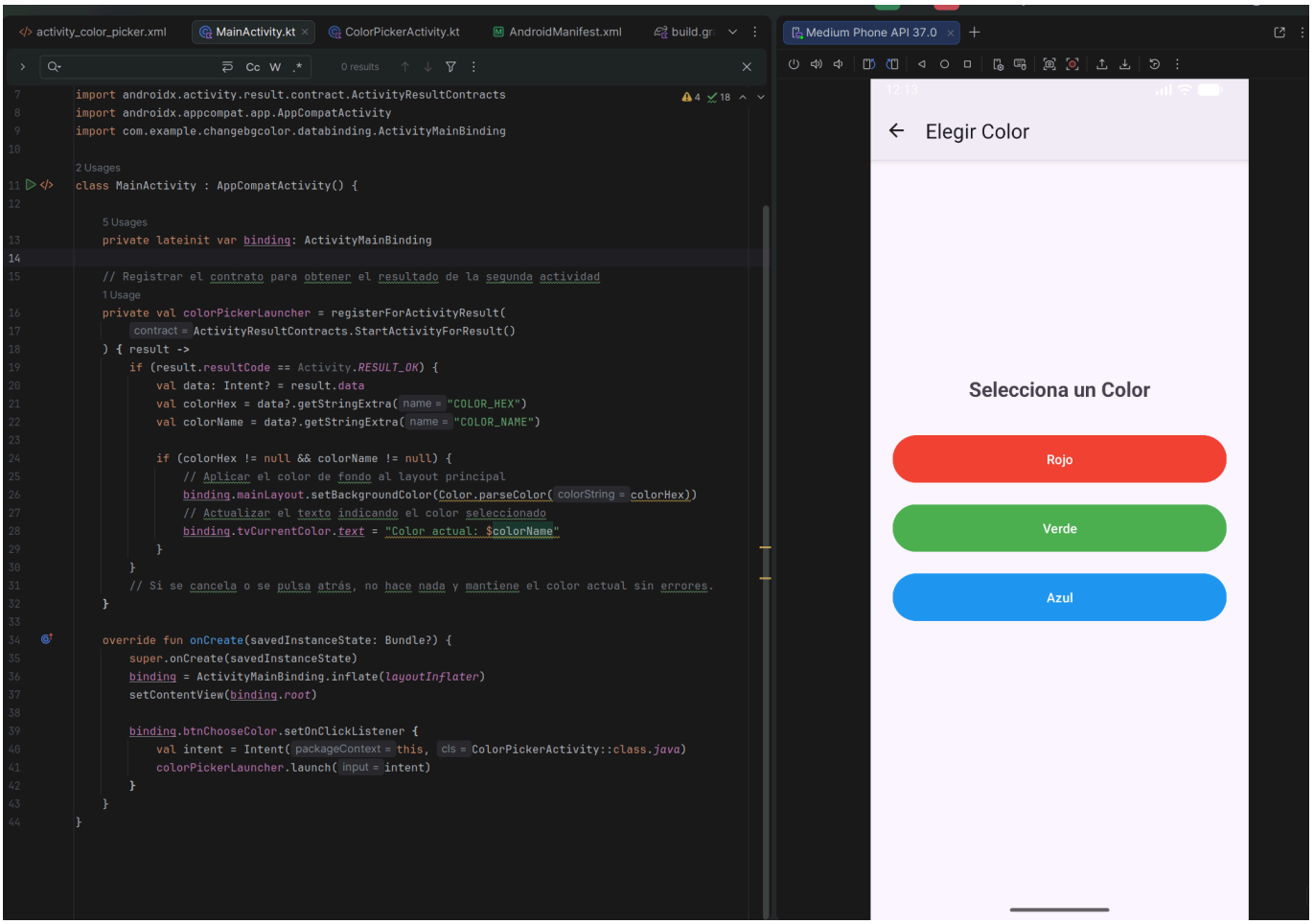
        <!-- Pantalla de Selección de Color -->
        <activity
            android:name=".ColorPickerActivity"
            android:exported="false"
            android:parentActivityName=".MainActivity" />

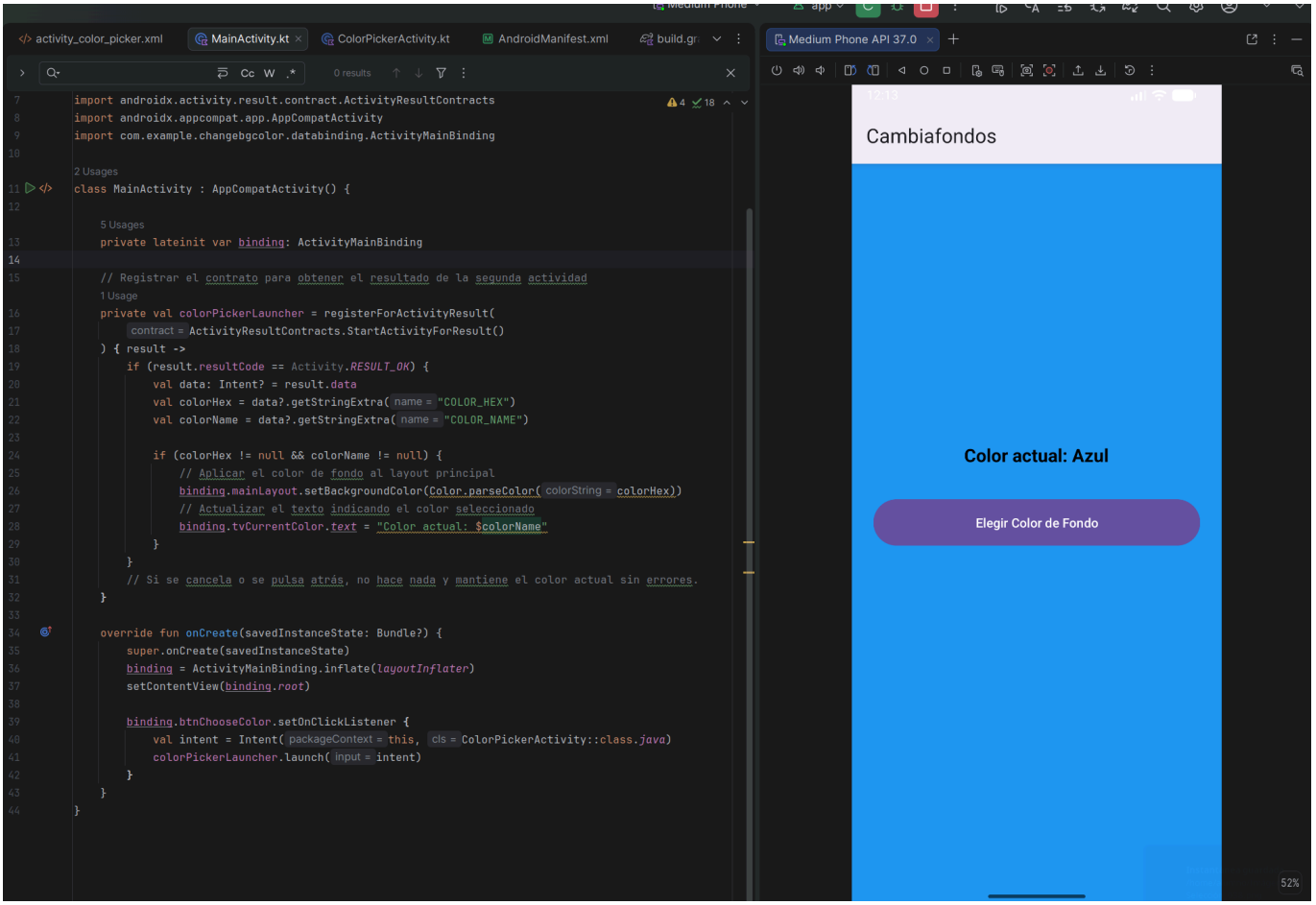
    </application>

</manifest>
```

Capturas de pantalla del código en funcionamiento


Medium Phone
app
ity_color_picker.xml
MainActivity.kt
ColorPickerActivity.kt
AndroidManifest.xml
build.gr
0 results
import androidx.activity.result.contract.ActivityResultContracts
import androidx.appcompat.app.AppCompatActivity
import com.example.changebgcolor.databinding.ActivityMainBinding
2 Usages
class MainActivity : AppCompatActivity() {
5 Usages
private lateinit var binding: ActivityMainBinding
// Registrar el contrato para obtener el resultado de la segunda actividad
1 Usage
private val colorPickerLauncher = registerForActivityResult(
contract = ActivityResultContracts.StartActivityForResult()
) { result ->
if (result.resultCode == Activity.RESULT_OK) {
val data: Intent? = result.data
val colorHex = data?.getStringExtra(name = "COLOR_HEX")
val colorName = data?.getStringExtra(name = "COLOR_NAME")
if (colorHex != null && colorName != null) {
// Aplicar el color de fondo al layout principal
binding.mainLayout.setBackgroundColor(Color.parseColor(colorString = colorHex))
// Actualizar el texto indicando el color seleccionado
binding.tvCurrentColor.text = "Color actual: $colorName"
}
}
// Si se cancela o se pulsa atrás, no hace nada y mantiene el color actual sin errores.
}
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
binding = ActivityMainBinding.inflate(layoutInflater)
setContentView(binding.root)
binding.btnChooseColor.setOnClickListener {
val intent = Intent(packageContext = this, cls = ColorPickerActivity::class.java)
colorPickerLauncher.launch(input = intent)
}
}
}
Medium Phone API 37.0
Cambiafondos
Color actual: Predeterminado
Elegir Color de Fondo


activity_color_picker.xml
MainActivity.kt
ColorPickerActivity.kt
AndroidManifest.xml
Medium Phone API 37.0
Elegir Color
Selecciona un Color
Rojo
Verde
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